

Problem: From the following table calculate GFR, TFR, ASFR_i, ASFR_f, GRR and NRR. Also given sex ratio at birth = 1.08 or male:female ratio = 52:48

Age	No. of Women	Total Birth	Survival Factor
15-19	4704852	148505	0.98
20-24	5037627	855994	0.977
25-29	4962900	1027579	0.974
30-34	3321030	501110	0.97
35-39	2798413	222705	0.966
40-44	2224489	61040	0.959
45-49	1678304	4934	0.948

Solution:

General Fertility Rate: GFR is defined as

$$GFR = \frac{\text{No. of live births that occur during a year}}{\text{Mid-year population of women aged 15-49}} \times 1000$$

In algebraic notations,

$$GFR = \frac{B}{W_{15-49}} \times 100$$

where,

B = No. of live births that occur during a year

W₁₅₋₄₉ = Mid-year population of women aged 15-49

Age-Specific Fertility Rate: ASFR_i is expressed symbolically,

$$ASFR_i = f_i = \frac{B_i}{W_i}$$

where, B_i = no. of births to women of i th age group

W_i = no. of women in the i th age group

f_i = Age-specific fertility rate of women in the i th age group

Total Fertility Rate: TFR for the population is given by

$$TFR = ACFR_{45-49} = 5 \sum_{i=15-19}^{45-49} f_i = 5 \sum_{i=15-19}^{45-49} ASFR_i$$

Age-specific Fertility Rate for Female Birth Only: It is defined as

$$ASFR_{f_i} = \frac{B_{f_i}}{W_i}$$

where, B_{f_i} = no. of female live births to mothers of i th age interval.

Gross Reproduction Rate: The formula for computing GRR is as follows:

$$GRR = 5 \sum_{i=15-19}^{45-49} \frac{B_{f_i}}{W_i} = 5 \sum_{i=1}^7 \frac{B_{f_i}}{W_i}$$

$$\text{OR, } GRR = TFR \left(\frac{B_f}{B_m + B_f} \right) = TFR \left(\frac{1}{1 + \text{Sex ratio at birth}} \right)$$

Net Reproduction Rate: The formula for calculating NRR:

$$NRR = \sum_{i=1}^7 f_i^d \left(\frac{5L_x}{L_0} \right)$$

where, f_i^d = Age-specific fertility rates based on only female births in the numerator

$\frac{5L_x}{L_0}$ = Survival rate base on the female life table

Interpretation of NRR:

1. $NRR=1$, means exact replacement or one daughter per woman.
2. $NRR<1$, denotes below-replacement fertility, where there are fewer daughters to succeed their mother's generation.
3. $NRR>1$, indicates above-replacement fertility - the future generation of potential mothers will be larger than the one that produced them.

The relevant calculation is given in the following table:

Age	No. of Women	Total Birth	Survival Factor	$ASFR_i$	Female Birth	$ASFR_{fi}$
15-19	4704852	148505	0.98	0.0315	71397	0.0151
20-24	5037627	855994	0.977	0.1699	411536	0.0816
25-29	4962900	1027579	0.974	0.2070	494028	0.0995
30-34	3321030	501110	0.97	0.1508	240918	0.0725
35-39	2798413	222705	0.966	0.0795	107070	0.0382
40-44	2224489	61040	0.959	0.0274	29346	0.0131
45-49	1678304	4934	0.948	0.0029	2372	0.0014
Total	24727615	2821867	6.774	0.6693	1356667	0.3218

Female Birth: Let, female birth = F

then male birth = $1.08 F$ $[\because \text{sex ratio at birth} = 1.08]$

So, total births = $F + 1.08 F = 2.08 F$

$$\Rightarrow F = \frac{\text{Total births}}{2.08}$$

From the table, $B = 2821867$, $\sum_{i=1}^7 ASFR_{fi} = 0.3218$

$W_{15-49} = 24727615$,

$$\sum_{i=1}^7 ASFR_i = 0.6693, \quad \sum_{i=1}^7 f_i^d \left(\frac{SL_i}{T_i} \right) = 0.3129$$

Therefore, $GFR = \frac{2821867}{24727615} = 114.118$

$TFR = 5 \times 0.6693 = 3.3469$

$GRR = 5 \times 0.3218 = 1.6091$

$NRR = 5 \times 0.3129 = 1.5647$

Comment: The value of GFR is 114.118, indicates approximately 114 births per 1000 women of reproductive age (ages 15-49). The population has a comparatively high birth rate among women of childbearing age. In the table, we compute ASFR_i and ASFR_f. A TFR of 3.3469 means on average, a woman is expected to give birth to about 3 or 4 children during her reproductive lifetime.

Since we get the value of $GRR = 1.6091$ which is greater than 1. So it indicates an increasing population size. The value of NRR of the above problem is 1.5647, which is greater than 1, so the data indicates above-replacement fertility, the future generation of potential mothers will be larger than the one that produced them.

Problem: The following table gives the population of a country for the year 2015, with the estimated number of births and deaths :

Age	Male		Female		Birth	
	Population	Death	Population	Death	Male	Female
0-4	442532	18623	434980	17308	-	-
5-9	419042	1809	416736	1709	-	-
10-14	393543	948	384616	1638	-	-
15-19	308269	1233	314056	1329	3578	3043
20-24	257852	1289	269340	1481	7293	6690
25-29	230629	1776	236187	1671	6775	6261
30-34	204188	1633	203477	1665	4233	4187
35-39	182270	1588	176534	1289	2999	2685
40-44	162509	1967	145037	1233	593	725
45-49	128784	2138	122946	1352	129	128
50-54	102971	1905	96589	1188	-	-
55-59	80717	2478	78311	1605	-	-
60-64	58899	3099	58142	1980	-	-
65-69	37797	2428	39049	2468	-	-
70+	45099	5981	48866	7175	-	-
Total	3055101	48895	3024866	45091	25600	24119

1. Calculate Crude death rate (CDR) of the total population for males and females.
2. Calculate GFR, ASFR, TFR and GRR.
3. Comment on your findings.

Solution: The CDR is the number of deaths in a given year per 1000 mid-year population i.e.

$$CDR = \frac{D}{P} \times 1000 \quad [CDR = \text{Crude Death Rate}]$$

where, D = no. of death in an area during a specified time

P = Mid-year population of that area

ASFR	ASFR (Female)	
-	-	This formula can be used separately for men and women.
-	-	
-	-	
0.0220	0.0106	The formulae of GFR, ASFR, TFR and GR are stated in the previous problem.
0.0519	0.0248	
0.0556	0.0269	
0.0413	0.0205	
0.0321	0.0152	
0.0090	0.0049	From the table, $D_m = 48895$, $P_m = 305510$ $D_f = 45091$, $P_f = 302486$ $W_{15-49} = 1467577$ [Using excel] $B = 49719$ $\sum_{i=1}^7 ASFR = 0.2143$ $\sum_{i=1}^7 ASFR_f = 0.1042$
0.0020	0.0010	
-	-	
-	-	
-	-	
-	-	
-	-	
-	-	
-	-	
0.2143	0.1042	

$$(1) CDR (\text{male}) = \frac{D_m}{P_m} \times 1000 = \frac{48895}{305510} \times 1000 = 16.004$$

$$CDR (\text{female}) = \frac{D_f}{P_f} \times 1000 = \frac{45091}{302486} \times 1000 = 14.907$$

$$(2) GFR = \frac{B}{W_{15-49}} \times 1000 = \frac{49719}{1467577} \times 1000 = 33.8782$$

ASFR_i and ASFR_{fi} have calculated in the table.

$$TFR = 5 \sum_{i=1}^7 ASFR_i = 5 \times 0.2143 = 1.0716$$

$$GRR = 5 \sum_{i=1}^7 ASFR_{fi} = 5 \times 0.1042 = 0.5212$$

(3) Comment on our findings: At first, we calculate crude Death Rate (CDR) for male and female. The values are 16.004 and 14.967 respectively. About 16 male deaths per 1000 male population per year. On the other hand, about 15 female deaths per 1000 female population per year. Male mortality is slightly higher than female mortality.

Then we calculate GFR = 33.8782, indicates approximately 34 births per 1000 women of reproductive ages. After that we compute ASFR_i and ASFR_{fi} in the table. A TFR of 1.0716 means on average, a woman is expected to give birth to about 1 child during her reproductive lifetime. GRR = 0.5212, indicates on average, a woman is expected to give birth to about 1 daughters during her reproductive lifetime.

Problem: From the following information find the singulate mean age at marriage (SMAM) for both male and female.

i	Age Group	Male never married s_i	Female never married s_i
1	10-14	100	93.8
2	15-19	93	43.4
3	20-24	61.5	11.2
4	25-29	20.6	3.4
5	30-34	5.3	1.0
6	35-39	1.4	0.6
7	40-44	1.0	0.3
8	45-49	1.1	0.2
9	50-54	0.2	0.1

Solution: The formula of SMAM is presented as follows

$$SMAM = \frac{N + 5 \sum_{i=1}^7 s_i - (50 \times \bar{s})}{100 - \bar{s}}$$

where, $N = 1000$

s_i = Percentage never married in each 5-years age group

$\bar{s} = \frac{1}{2}(s_8 + s_9)$ = Percentage never married at age 50 or in the final age group, often estimated from ages 45-49 and 50-54.

SMAM = singulate Mean Age at marriage.

SMAM for Male: By using excel,

$$N = 1000$$

$$\sum_{i=1}^7 S_i = 326$$

$$\bar{S} = \frac{1}{2}(S_8 + S_9) = \frac{1}{2}(4 + 0.3) = 2.15$$

$$\begin{aligned}\therefore \text{SMAM} &= \frac{1000 + 5 \times 326 - (50 \times 2.15)}{100 - 2.15} \\ &= 25.7792\end{aligned}$$

SMAM for female:

$$N = 1000$$

$$\sum_{i=1}^7 S_i = 185$$

$$\bar{S} = 0.3$$

$$\begin{aligned}\text{SMAM} &= \frac{1000 + 5 \times 185 - (50 \times 0.3)}{100 - 0.3} \\ &= 19.1574\end{aligned}$$

Comment: Men marry for the first time at about 25.8 years on average. On the other hand, Women marry for the first time at about 19.2 years on average.

Men tend to marry considerably later than Women in the given population studied. Women are entering marriage at a relatively young age compared to men.

Problem: For Bangladesh with the following information estimate growth rate of population:

$$\text{Population 1991} = P_1 = 89.9$$

$$\text{Population 2001} = P_2 = 111.45$$

Hence find out the doubling time.

Solution: For linear growth function,

$$r = \frac{P_2 - P_1}{P_1 t}$$

where,

P_1 = population in 1991

P_2 = Population in 2001

t = time period

r = growth rate

$$\begin{aligned}\therefore r &= \frac{111.45 - 89.9}{10 \times 89.9} \\ &= 0.023971\end{aligned}$$

For geometric growth function,

$$\begin{aligned}r &= \left(\frac{P_2}{P_1} \right)^{1/t} - 1 \\ &= \left(\frac{111.45}{89.9} \right)^{1/10} - 1 = 0.02172\end{aligned}$$

For exponential growth function

$$r = \frac{1}{t} \ln \left(\frac{P_2}{P_1} \right)$$

$$r = \frac{1}{10} \ln \left(\frac{111.45}{89.9} \right)$$

$$= 0.021488$$

Calculating the time to double the population size:

For linear growth function,

$$t = \frac{1}{r} = \frac{1}{0.023971} \\ = 41.71694 \text{ years}$$

For geometric growth function,

$$t = \frac{\ln 2}{\ln(1+r)} = \frac{\ln 2}{\ln(1+0.02172)} \\ = 32.25769 \text{ years}$$

For exponential growth function,

$$t = \frac{\ln 2}{r} = \frac{\ln 2}{0.021488} \\ = 32.25769 \text{ years}$$

Comment: Firstly, we compute the growth rate for linear, geometric and exponential growth function. The values are 2.40%, 2.17% and 2.15% respectively. Secondly, we calculate the time to double the population size. The required time to double the population 41.72, 32.26 and 32.26 years for the above cases.